

HARSH SHAH

Principal Solution Architect | PX Platform Strategy | Target-State Architecture & Platform Modernization
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EXECUTIVE SUMMARY

Principal solution architect and engineering leader with 19 years of experience helping complex enterprise platforms evolve from tightly coupled legacy ecosystems into modular, API-driven, cloud-ready architectures. Current work centers on the PX / Planning Solutions platform: shaping how Fact Finding and Planning capabilities move toward cleaner domain boundaries, reusable platform services, experience APIs, domain APIs, and data orchestration. Brings deep past experience across financial services, insurance, retirement, industrial platforms, AWS/Azure modernization, event-driven integration, engineering governance, and executive technology advisory. Coordinates PX architecture through SART and enterprise governance channels to keep platform direction aligned with enterprise patterns, security standards, data-access guardrails, operational readiness, and the practical needs of planning teams.

PX PLATFORM TARGET-STATE ARCHITECTURE

- Authored the PX Platform target-state architecture and modernization roadmap, defining how planning capabilities evolve from a tightly coupled ecosystem into independently scalable domains connected through APIs and reusable platform services.
- Serve as the Planning Solutions architect within SART, helping shape PX platform direction by aligning architecture decisions with enterprise patterns, security guardrails, data-access standards, operational readiness, and long-term technology strategy.
- Built the modernization strategy around domain boundaries, API ownership, data separation, and incremental migration paths so planning capabilities can evolve without disrupting advisor experiences.
- Worked across enterprise architecture, security, platform engineering, and product teams to ensure PX capabilities followed enterprise patterns while staying grounded in the needs of planning teams and advisor experiences.
- Defined integration direction across PX Platform, Advice Workspace, Jump, Data Import Manager, and client/account data capabilities to support reusable planning services and future-state interoperability.
- Helped guide migration planning for Advice Workspace / ORA, using dual-write thinking, feature toggles, rollback options, and data-access considerations to reduce risk while teams move from legacy capabilities to the new platform model.

WHAT I BRING TO THE TABLE

Target-State Platform Strategy: PX Platform TSA | Modularization | L1/L2 Architecture | Roadmaps | Interim-to-Target Transition | Architecture Governance

SART & Enterprise Governance: Enterprise Patterns | Security Standards | Architecture Reviews | Data-access Guardrails | Operational Readiness | Governance Coordination

Platform Architecture: Domain APIs | Experience APIs | Data Orchestration | Event-Driven Architecture | Service Boundaries | Reusable Platform Capabilities

Planning Domain: Fact Finding | Planning Experience | ORA Platform | ALAI | Data Import Manager | Advisor Joint Work | Co-Client Planning | Plan Progress

Leadership & Platform Strategy: CTO Advisory | Architecture Reviews | Product-Engineering Alignment | Technical Strategy | Governance | Production Readiness

PLATFORM MODERNIZATION HIGHLIGHTS

- **Retirement platform transformation:** Partnered with a retirement-services CTO to turn a fragmented ecosystem into a unified digital platform for members, advisors, sponsors, and TPAs; plan onboarding improved from 25 days to 32 hours and operational efficiency improved 64%.

- **Insurance servicing modernization:** Moved NM servicing capabilities away from brittle point-to-point integrations into an event-driven AWS/Kafka model where issuance, policy updates, reporting, and commissions could react independently; servicing turnaround dropped from 24 days to 14 days.
- **Komatsu digital platform:** Directed architecture for a next-generation fleet-management platform, bringing machine, distributor, and customer experiences together through reusable APIs and shared platform foundations.
- **Enterprise document platform:** Built a modern document platform that replaced microfilm and microfiche across Life Insurance areas while creating CI/CD standards, onboarding practices, and a path to retire legacy infrastructure.

PROFESSIONAL EXPERIENCE

Northwestern Mutual | Apr 2025 – Present

Solution Architect – Planning Solutions / PX Platform

- Serve as the Planning Solutions architect in SART and enterprise architecture forums, partnering with Product, Engineering, Compliance, Enterprise Architecture, and downstream platform teams to keep PX platform decisions practical, scalable, secure, and grounded in how teams actually build.
- Coordinate enterprise pattern alignment for PX capabilities across domain APIs, gRPC integrations, data access, security and entitlement considerations, rollout strategy, and operational readiness before major platform commitments are locked.
- Lead architecture strategy across PX target-state capabilities including ORA Platform, Active Listening AI, Data Import Manager, Plan Progress, multi-advisor collaboration, co-client planning, and gRPC-based planning service integrations.
- Shape ORA Platform capabilities that capture client objectives, advisor recommendations, achievements, and AI-assisted objective classification as reusable planning-domain capabilities across advisor experiences.
- Integrate Active Listening AI as a separate GenAI/LLM platform capability that extracts structured financial facts from advisor-client conversations and connects those facts into advisor workflows and downstream planning experiences.
- Architect Data Import Manager for multi-import/export of client and household demographics and financial facts, supported by automated fact resolution and reusable Domain API consumption.
- Guide practical implementation patterns for migration, rollback, feature toggles, service boundaries, entitlement-sensitive collaboration, and operational readiness across planning capabilities.

Infosys Limited | Oct 2007 – Apr 2025

Director, Digital Products / Principal Architect / Solution Architect

Customer-facing engineering and architecture leader across Fortune 500 technology programs in financial services, insurance, retirement, industrial manufacturing, mortgage, and digital commerce; progressed from hands-on engineering to principal architecture, engineering management, and director-level platform leadership.

Komatsu — Member Experience Portal | Apr 2024 – Apr 2025 | Director, Digital Products

- Led architecture and platform direction for a fleet-management platform serving distributors and machine owners across Construction, Mining, and Forestry, aligning product, engineering, data, and business stakeholders around a market-driven roadmap.
- Directed a 24+ person cross-functional team and established reusable APIs, integration patterns, and operationally ready architecture foundations to support after-sales parts growth.

Leading Retirement Record Keeper | Apr 2022 – Apr 2024 | Principal Architect, Digital Products

- Worked directly with the CTO to define a retirement services transformation strategy and guide Azure-based platform execution across member, advisor, plan sponsor, and TPA experiences.
- Ran Architecture Review Board governance and secured formal architecture approval for a regulated platform modernization satisfying business, operational, and compliance requirements.
- Delivered capabilities that improved plan onboarding from 25 days to 32 hours, supported a micro-market segment that grew approximately 45% year over year, and improved operational efficiency by

64%.

Northwestern Mutual / Infosys — Insurance Servicing Modernization | Apr 2021 – Mar 2022 | Solution Architect & Engineering Manager

- Modernized Life, Disability Income, and Long-Term Care servicing from point-to-point integrations to an event-driven AWS model using Kafka, EKS, Lambda, RDS, Aurora, S3, and Redis.
- Created an event hub for issuance, policy updates, reporting, and commissions to react independently; servicing turnaround dropped from 24 days to 14 days and operational efficiency improved 25%.

Earlier Programs | 2007 – 2021 | Solution Architect, Engineering Manager, Technical Lead, Senior Engineer

- Designed an Azure Container Services platform connecting Marketing, Servicing, and Communications systems at Movement Mortgage; mortgage issuance grew 150% the following year.
- Built enterprise document management capabilities replacing microfilm and microfiche across Life Insurance business areas; established CI/CD standards, managed onboarding, and helped decommission legacy infrastructure.
- Led architecture and engineering across Contracts Production, Sales Illustration, Enterprise Electronic Forms, Re-Insurance, and government/business portal systems, building a long runway of experience modernizing regulated, integration-heavy platforms.

SELECTED ARCHITECTURE PORTFOLIO

- **Event-Driven, AI-Assisted Claims Platform:** Reference architecture for regulated insurance claims using Kafka, event sourcing, CQRS, bounded contexts, Claims Domain APIs, fraud/STP decisioning, payment saga patterns, and human-in-the-loop LLM agents.
- **Machine Telemetry Platform:** Reference architecture for industrial telemetry using Palantir Foundry, Kafka streaming, machine domain APIs, Adobe Commerce integration, NBO/NBA decisioning, observability, FinOps, and security controls.
- **Algo Trading App:** Working paper-trading momentum engine for stocks, ETFs, and options using Schwab API integration, a tiered 9-signal decision model, AWS deployment, and risk controls designed to keep automation disciplined before real-money exposure.

CERTIFICATIONS, THOUGHT LEADERSHIP, EDUCATION & COMMUNITY

- AWS Certified Solutions Architect (2020). Published Infosys whitepapers on Microapp Architecture with Domain-Driven Design and Hybrid vs. Native Mobile App Development.
- B.E. Electrical Engineering, Birla Vishwakarma Mahavidyalaya – Sardar Patel University, India.
- Facilities Secretary, Jain Society of Metropolitan Chicago; Girls in Tech Program; EE&T Pathway Leadership Mentor; Northwestern Mutual HiTech STEM Outreach; Kids Reading Program.